

Incident Response & Containment Report

– Parrot OS

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Environment: Parrot OS VM (VirtualBox)

Date: [Insert Date]

1. Objective

Demonstrate containment procedures for a compromised VM, including network isolation, firewall enforcement, evidence preservation, service shutdown, and network segmentation.

2. Network Isolation

Step: Identify active interface

```
ip addr
```

- **Interface used:** `enp0s1`
- **Screenshot:** [Insert before isolation]

Step: Isolate interface

```
sudo ip link set enp0s1 down  
ping 8.8.8.8
```

- **Expected:** `Network is unreachable` → confirms isolation
- **Screenshot:** [Insert ping failure]

Step: Reconnect interface

```
sudo ip link set enp0s1 up  
ping 8.8.8.8
```

- **Expected:** Ping successful → network restored
 - **Screenshot:** [Insert successful ping]
-

3. Firewall Implementation

```
sudo ufw enable  
sudo ufw default deny incoming  
sudo ufw status verbose
```

- **Purpose:** Block inbound traffic for containment
- **Screenshot:** [Insert ufw status]

Optional: Block specific IPs

```
sudo ufw deny from 192.168.1.50
```

4. Evidence Preservation

- **File System Artifacts**

```
mkdir -p ~/Evidence  
md5sum /path/to/suspicious/file >> ~/Evidence/file_hashes.txt  
sha1sum /path/to/suspicious/file >> ~/Evidence/file_hashes.txt
```

- **Screenshot:** [Insert Evidence folder contents]

- **Network Traffic Capture**

```
sudo tcpdump -i enp0s1 -w ~/Evidence/network_capture.pcap
```

- **Screenshot:** [Insert tcpdump capture]
- **Memory Dump (Optional)**

```
sudo dd if=/dev/mem of=~/Evidence/memdump.raw bs=1M
```

- **Screenshot:** [Insert memory dump]
-

5. Service Shutdown

Stop potentially compromised services:

```
sudo systemctl stop apache2
sudo systemctl stop ssh
sudo systemctl status apache2
sudo systemctl status ssh
```

- **Screenshot:** [Insert stopped services]
-

6. VirtualBox Network Segmentation (Optional)

- Change VM network adapter to **Internal Network** or **Host-only Adapter** to physically isolate the VM.
- **Screenshot:** [Insert VirtualBox network settings]

7. Summary Table

Step	Action	Evidence
Network Isolation	Interface down (<code>enp0s1</code>)	Ping failure screenshot
Firewall	<code>ufw default deny incoming</code>	Firewall status screenshot
Evidence Preservation	Hashes, tcpdump, memory dump	Evidence folder screenshot
Service Shutdown	Stop apache2/ssh	<code>systemctl status</code> screenshot
Network Segmentation	Internal Network / Host-only	VirtualBox settings screenshot

```
[user@parrot]~[~]
→ $ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 52:54:00:08:cf:dc brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.30/24 brd 192.168.1.255 scope global dynamic noprefixroute enp0s1
        valid_lft 2462sec preferred_lft 2462sec
    inet6 2600:480a:3393:6900::2e4b/128 scope global dynamic noprefixroute
        valid_lft 600173sec preferred_lft 254573sec
    inet6 2600:480a:3393:6900:f7a4:966c:388f:8715/64 scope global dynamic noprefixroute
        valid_lft 301sec preferred_lft 301sec
    inet6 fe80::b6a7:f58d:b0c7:1994/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

```
[user@parrot]-[~]  
$ ip addr show enp0s1  
2: enp0s1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen  
link/ether 52:54:00:08:cf:dc brd ff:ff:ff:ff:ff:ff  
inet 192.168.1.30/24 brd 192.168.1.255 scope global dynamic noprefixroute enp0s1  
    valid_lft 3485sec preferred_lft 3485sec  
inet6 2600:480a:3393:6900::2e4b/128 scope global dynamic noprefixroute  
    valid_lft 597956sec preferred_lft 252356sec  
inet6 2600:480a:3393:6900:f7a4:966c:388f:8715/64 scope global dynamic noprefixroute  
    valid_lft 301sec preferred_lft 301sec  
inet6 fe80::b6a7:f58d:b0c7:1994/64 scope link noprefixroute  
    valid_lft forever preferred_lft forever
```

```
[user@parrot]-[~]  
$ sudo ip link set enp0s1 down  
[user@parrot]-[~]  
$ ping 8.8.8.8  
ping: connect: Network is unreachable  
[x]-[user@parrot]-[~]  
$ sudo ip link set enp0s1 up  
[user@parrot]-[~]  
$ ping 8.8.8.8  
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
```

```
64 bytes from 8.8.8.8: icmp_seq=1 ttl=116 time=34.8 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=116 time=22.8 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=116 time=18.5 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=116 time=22.0 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=116 time=25.0 ms
64 bytes from 8.8.8.8: icmp_seq=6 ttl=116 time=26.3 ms
64 bytes from 8.8.8.8: icmp_seq=7 ttl=116 time=27.8 ms
64 bytes from 8.8.8.8: icmp_seq=8 ttl=116 time=16.7 ms
64 bytes from 8.8.8.8: icmp_seq=9 ttl=116 time=13.2 ms
64 bytes from 8.8.8.8: icmp_seq=10 ttl=116 time=24.1 ms
64 bytes from 8.8.8.8: icmp_seq=11 ttl=116 time=26.9 ms
64 bytes from 8.8.8.8: icmp_seq=12 ttl=116 time=28.1 ms
64 bytes from 8.8.8.8: icmp_seq=13 ttl=116 time=23.5 ms
64 bytes from 8.8.8.8: icmp_seq=14 ttl=116 time=24.2 ms
```

```
--- 8.8.8.8 ping statistics ---
```

```
457 packets transmitted, 451 received, 1.31291% packet loss, time 457744ms
```

```
rtt min/avg/max/mdev = 12.352/27.730/134.969/13.995 ms
```